

Xinyue YU (Christina)

+1 (312) 285 6305 | yu1619@purdue.edu | West Lafayette, IN, U.S.
christinaxinyueyu.github.io

Curriculum Vitae
Updated in Aug. 2026

EDUCATION

Purdue University, Edwardson School of Industrial Engineering

PhD, Ross Fellowship Awardee

West Lafayette, IN, USA

Sept. 2025 – May 2030 (expected)

- Advised by Dr. Nan Kong and Dr. Yuehwern Yih
- Relevant courses:** Engineering Economic Analysis, Design and Analysis of Experiments, Human Factors in Engineering, Integer Programming

The University of Hong Kong

Bachelor of Science (Major in Mathematics), Lee Shau Kee Scholar, First Class Honors

Hong Kong S.A.R.

Sept. 2021 – June 2025

McGill University

Exchange Student

Montreal, Canada

Jan. 2023 – Apr. 2023

RESEARCH INTERESTS

Operations Research, Resource Allocation, Healthcare Operations, Bilevel Mixed Integer Programming, Algorithm Design

RESEARCH EXPERIENCE

A Single-Leader Multi-Follower BMIP Approach to Government Incentive Optimization

Supervised by Dr. Nan Kong, Purdue University

Sept. 2025 – Present

- Assisted in formulating the optimal subsidy allocation problem with a single-leader–multiple-follower bilevel integer programming model for the resultant Stackelberg game, where the government acts as the leader and each health system responds optimally and independently as multiple followers serving mutually disjoint areas, assisted in designing a decomposition-based algorithm that decomposes the formulated bilevel integer program into single-leader–single-follower subproblems and conducting instance testing

Optimal Control for Multi-Agent System (Final Year Project Research)

Supervised by Dr. Tak Kwong Wong, The University of Hong Kong

Aug. 2024 – May 2025

- Referred to literature in linear quadratic discrete choice problems, explored further on realistic applications in optimal exits design for theaters and auditoriums
- Studied in the collective choice model (CCM) introduced in literature of linear quadratic mean field game (LQ-MFG), compared its strengths and drawbacks with other formulations

AI Prototype for Mental and Resources Support for Care Givers in Hong Kong

Worked with Agewhale, a startup in care givers support in Hong Kong

Aug. 2024 – July 2025

- Cooperated with Agewhale to develop technical solutions for family carers to navigate and coordinate elder care approaches and resources in Hong Kong, offered accurate and updated information on related inquiries by the AI model
- Supported in data collection, sorting, and filtering, formulated the customer journey of users and the database, optimized the answers provided by the AI model by clearing redundant data and rephrasing wordings

Basics of Optimal Control and Simple Real-World Applications

Supervised by Prof. Tak Kwong Wong, The University of Hong Kong

June 2023 – Aug. 2023

- Applied optimal control on the simulation of the scenario for multiple audience after a movie, tried to minimize the time for all agents to depart from the theater with no clash and the velocity is proportional to the average distance with other agents, coded in MATLAB to demonstrate the trajectory and velocity of every agent in 1-agent case and 2-agent case

Supervised Learning in Low Light Signal Detection Biomedical Experiments

Supervised by Dr. Vyacheslav Kalchenko, Weizmann Institute of Science (Israel)

Aug. 2022

- Cowrote the report *Making the Invisible Visible: Regression and Artificial Neural Networks in Low Light Signal Detection*
- Diagnosed the overuse of live animals in low-light detection biomedical experiments and developed ML approaches
- Applied supervised learning to train a model on the imitation of human visual recognition and animal reactions

HONORS, AWARDS & MEMBERSHIP

The University of Hong Kong

Lee Shau Kee Scholarship , full tuition scholarship awarded by HKU & Lee Shau Kee Foundation	Sept. 2021 – July 2025
Rosita King Ho Scholarship , exchange scholarship awarded by HKU & Rosita King Ho Foundation	May 2023
Dean's Honor List , awarded by HKU Faculty of Science	Sept. 2021 – May 2022
Young Scientist Scheme Member in HKU Faculty of Science	Sept. 2021 – May 2023

Croucher Foundation

Full sponsorship in the <u>International Science Summer Institute</u> hosted in Weizmann Institute of Science (Israel)	July 2022
---	-----------

Chinese Mathematical Society

Provincial First Prize in China National High School Mathematics Olympiad	Oct. 2020
--	-----------

TEACHING EXPERIENCE

Teaching Assistant, The University of Hong Kong

MATH2822 Mathematical Methods for Actuarial Science II	Jan. 2025 – May 2025 (assigned)
MATH1821 Mathematical Methods for Actuarial Science I	Sept. 2024 – Dec. 2024
COMP1117 Computer Programming (Python)	Sept. 2023 – Dec. 2023

WORK EXPERIENCE

China International Capital Corporation

Shanghai, China

Equity Research Intern

May 2024 – Aug. 2024

- Researched on global energy drinks market for competitive landscape change, product features, and consumer preferences
- Studied in Japan beer market history, liquor tax policies, major market players, and market portfolio change

Petrel Capital Greater China Fund

Hong Kong S.A.R.

Equity Research Intern

June 2023 – Aug. 2023

- Covered the materials and consumer sector, studied the main players, their market shares, and the change of shares
- Deep dived into the sportswear industry, analyzed the manufacturing and destocking schedule of top-10 market players

Yunnan Baiyao Group Pharmaceutical E-Commerce Co., Ltd.

Kunming, China

Data Lake Project Intern

May 2022 – July 2022

- Merged databases based on the business model using SQL, optimized the structure of dictionaries to improve efficiency in looking up

LEADERSHIP, ACTIVITIES & VOLUNTEERING EXPERIENCE

The University of Hong Kong and Faculty of Science

Hong Kong S.A.R.

Student Ambassador

Sept. 2021 – Present

The University Business Consulting Club

Hong Kong S.A.R.

Vice President & Leader of Training Department

Apr. 2023 – Apr. 2024

- Managed the weekly training on financial market overview, case interview training, and the pharmaceutical industry
- Administered freshman orientation camp organizing, club members recruitment, and business competition hosting

PwC's Metaverse Challenge 2023

Hong Kong S.A.R.

Team Member of Audience Favorite

July 2023

- Designed the architecture and optimized metaverse onboarding experience for a company after acquisition in Sandbox

Sunflower Program, Shenzhen Shiyu Children Foundation

Kunming & Shenzhen, China

New Media Group Member and Social Work Volunteer (140 hrs in Kunming, 20 hrs in Shenzhen)

July 2018 – Present

- Volunteered in ward caring enterprise hosted in Shenzhen and Kunming for children suffering from tumors and cancers

SKILLS & CERTIFICATES

Languages: English, Mandarin (native), Cantonese (basic)

Interests: Art curating, movies and TV series, music (R&B, popular), calligraphy, violin

Certificates: CFA ESG Investing Certificate, Society of Actuaries P Exam Passed